

Example 12.22 Following are the runs scored by a batsman in last 20 test matches: 96, 70, 100, 96, 81, 84, 90, 89, 63, 90, 34, 75, 39, 82, 85, 86, 76, 64, 67, and 88. Calculate the quartile deviation and coefficient of quartile deviation.

Solution: At first, arranging the data in ascending order:

34, 39, 63, 64, 67, 70, 75, 76, 81, 82, 84, 85, 86, 88, 89, 90, 90, 96, 96, 100

First Quartile (Q_1): The calculation of first quartile is shown below:

$$Q_i = \frac{i(n+1)}{4} \text{th observation}$$

$$\therefore Q_1 = \frac{1(20+1)}{4} \text{th observation} = 5.25^{\text{th}} \text{ observation}$$

Since 5.25^{th} observations lies between 5^{th} and 6^{th} value in the ordered group, or midway between 67 and 70. Therefore,

$$Q_1 = 67 + 0.25(70 - 67) = 67 + 0.75 = 67.75$$

Third Quartile (Q_3): The formulation for the calculation of third quartile is given below:

$$Q_i = \frac{i(n+1)}{4} \text{th observation}$$

$$\therefore Q_3 = \frac{3(20+1)}{4} \text{th observation} = 15.75^{\text{th}} \text{ observation}$$

Since 15.75^{th} observations lies between 15^{th} and 16^{th} value in the ordered group, or midway between 89 and 90. Therefore,

$$Q_3 = 89 + 0.75(90 - 89) = 89 + 0.75 = 89.75$$

By putting the values into the formulas of quartile deviation and coefficient of quartile deviation we get:

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2} = \frac{89.75 - 67.75}{2} = 11$$

$$\text{Coefficient of Quartile Deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100 = \frac{89.75 - 67.75}{89.75 + 67.75} \times 100 = 6.89\%$$

Example 12.23 Following are the observations showing the age of 50 employees working in a whole sale centre. Find the quartile deviation and coefficient of quartile deviation.

Age in years	40 – 44	45 – 49	50 – 54	55 – 59	60 – 64	65 – 69
Employees	4	7	14	11	8	6